Problem Set 2

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Data Scientists use a variety of tools that revolve around technology and theory! Here are a few of those tools.

1 Statistical Programming Languages

1.1 Complied

- R (Generally the one we'll use most in class)
- Python
- Julia

1.2 Scripted

- Stata
- SAS
- SPSS
- MAtlab
- JavaScript (occasionally)

2 Web Scraping

Using an application program interface (API) to download data and extract them onto our programs.

2.1 APIs

- Employed by many notable web companies
- Companies can employ to guard their data
- ex: LinkedIn limits the information that a person can scrape

2.2 Parsing

- Data download by yourself
- IP related drawbacks
- Might get blocked from frequency

3 Handling Large Data Sets

Sometimes (most of the time), being a Data Scientist requires us to deal with massive data sets that involve cleaning, organizing, and restructuring what were given.

- Most computers do not have enough RAM to open large data sets
- You may be able to split data into more manageable chunks.
- Can use Resilient Distributed Datasets (RDDs)
- Coding in a Structured Query Language (SQL) can help subset and merge moderately sized datasets

4 Visualization

Being able to visualize our research is one of the most important skills to have as a Data Scientist.

- Allows humans to see how data can look like
- ggplot2 in R has a large user following
- matplotlib in Python can help develop some visuals as well
- Plits.jl on Julia is intuitive for its ability to use other packages as a backend
- Tabealu (we wont be using this in our coursework)

5 Modeling

- Answering predictions based on data
- Utilize data to optimize processes
- Explain behaviors through data